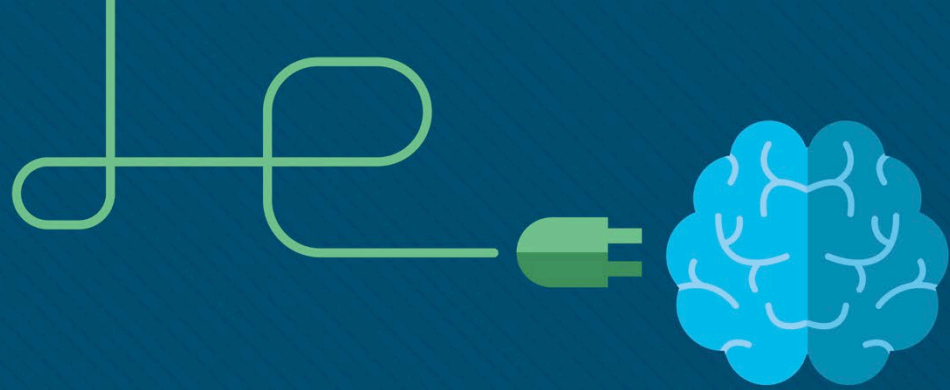




# Week 7

RH124 Chapter 13.5. Locate Files on the System



## Week 3.1 - Objectives

- Students should be thoroughly familiar with the Filesystem Hierarchy Standard (FHS), including typical file locations and directory classifications.
- **Key Knowledge Areas:**
  - Using find to locate and act on files based on type, size, or time.
  - Understand the correct locations of files under the FHS
  - Find files and commands on a Linux system.
  - Know the location and purpose of important file and directories as defined in the FHS.

# File Searching

# 1 Filesystem Hierarchy Standard

- The **Filesystem Hierarchy Standard (FHS)** is an agreement to standardize the names and locations of directories and their content for use within most Linux filesystems. It helps to know what directories to expect to find, and what files one should expect to find in them. More importantly, it allows programmers to write programs that will be able to work across a wide variety of systems that conform to this standard.
- The Linux file structure is best visualized as an upside-down tree, with directories and files branching out from the top-level root / directory. While the actual standard details many more directories than listed below, the image and table highlight some of the most important ones to know.

# 1 Filesystem Hierarchy Standard (Cont'd)

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- The Linux file structure is best visualized as an upside-down tree, with directories and files branching out from the top-level root `/` directory. While the actual standard details many more directories than listed below, the image and table on the next few slides highlight some of the most important ones to know.

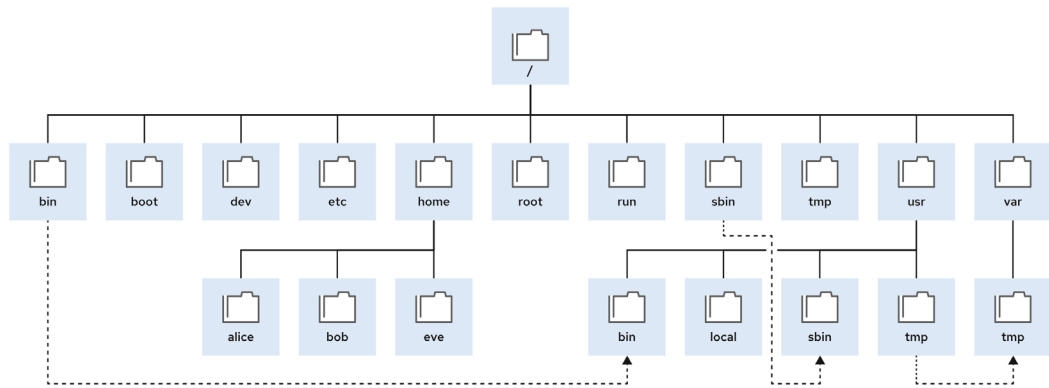


Figure 3.1: Significant file-system directories in Red Hat Enterprise Linux 9

## 5.1 Filesystem Hierarchy Standard (Cont'd)

| Dir name         | Purpose                                                    |
|------------------|------------------------------------------------------------|
| /                | The root of the primary filesystem hierarchy               |
| /bin             | Contains essential user binary executables                 |
| /boot            | Contains the kernel and bootloader files.                  |
| /dev             | Populated with files representing attached devices.        |
| /etc             | Configuration files specific to the host.                  |
| /home            | Common location for user home directories.                 |
| /lib and<br>/mnt | Essential libraries to support /bin and /sbin executables. |
| /opt             | Optional third-party add-on software.                      |
| /root            | Home directory for the root user.                          |
| /run             | Contain volatile data that changes at runtime.             |
| /sbin            | Contains system or administrative binary executables.      |

## 5.1 Filesystem Hierarchy Standard (Cont'd)

| Dir name                  | Purpose                                                                                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>/srv</code>         | May contain data provided by system services.                                                                                                                                             |
| <code>/sys</code>         | Hold files related to the kernel to export information about kernel objects                                                                                                               |
| <code>/tmp</code>         | Location for creating temporary files.                                                                                                                                                    |
| <code>/usr</code>         | The root of the secondary filesystem hierarchy.                                                                                                                                           |
| <code>/usr/bin</code>     | Non-vital system or administrative executables.                                                                                                                                           |
| <code>/usr/include</code> | Header files for compiling C-based software.                                                                                                                                              |
| <code>/usr/lib</code>     | Shared libraries to support <code>/usr/bin</code> and <code>/usr/sbin</code>                                                                                                              |
| <code>/usr/local</code>   | The root of the third filesystem hierarchy for local software.                                                                                                                            |
| <code>/usr/sbin</code>    | Non-vital system or administrative executables.                                                                                                                                           |
| <code>/usr/share</code>   | Location for architecturally-independent data files.                                                                                                                                      |
| <code>/var</code>         | Contains shareable filesystem eg <code>/var/www</code> or <code>/var/ftp</code> , system logs, audit journal entries, user cron tables and mailboxes. Also contain runtime process locks. |

## 2 Finding Files and Commands

- There are a few commands available for searching files and commands from the CLI. Both the *locate* and *find* commands are useful for searching for a file within the filesystem. While both commands perform similar tasks, each does so by using a different technique, with its own distinct advantages and disadvantages.
- The *locate* command can be described as fast, but always potentially out-of-date. Its speed comes from the fact that the *locate* command searches a database that contains the location of the files on the filesystem, but that database needs to be updated to be accurate.

```
locate [OPTION]... PATTERN...
```

- In its simplest form, the *locate* command takes a search string as argument

```
sysadmin@localhost:~$ locate passwd
/etc/passwd
/etc/passwd-
/etc/pam.d/chpasswd
/etc/pam.d/passwd
/etc/security/opasswd
/usr/bin/gpasswd
/usr/bin/passwd
/usr/include/rpcsvc/yppasswd.h
/usr/include/rpcsvc/yppasswd.x
/usr/lib/tmpfiles.d/passwd.conf
/usr/sbin/chgpasswd
/usr/sbin/chpasswd
/usr/sbin/update-passwd
/usr/share/base-passwd
/usr/share/base-passwd/group.master
/usr/share/base-passwd/passwd.master
```

## 2 Finding Files and Commands (Cont'd)

- Few things to consider when using the *locate* command:
  - The command only return results that the current user has access to
  - The search string can be anywhere in the file name
  - The search string is case sensitive (use the *-i* option for case insensitive search)
  - You can use the *-r* option to use regular expressions in the search pattern.
- The accuracy of the search results returned by the *locate* command is dependent on a database. This database can be manually updated by an administrator using the *updatedb* command which is also automatically run every day through the system scheduling service *cron*.
- Other things to take note about the use of the database by the *locate* command
  - The command is able to work very quickly as compared to the *find* command since it does not need to conduct the search in real time.
  - The database may be outdated if files/directories are removed from the file system
  - The */etc/updatedb.conf* can be edited to specify key terms in file names that should not be added to the database.

## 2 Finding Files and Commands (Cont'd)

1. A new file named *lostfile* is created but isn't found by the *locate* command
2. After the database is updated by executing the *updatedb* command, the *locate* command can find the *lostfile* file
3. After the *lostfile* file has been deleted, the *locate* command will still report that the file exists until the next successful run of the *updatedb* command

```
sysadmin@localhost:~$ touch lostfile
sysadmin@localhost:~$ locate lostfile
sysadmin@localhost:~$ sudo updatedb
[sudo] password for sysadmin:
sysadmin@localhost:~$ locate lostfile
/home/sysadmin/lostfile
sysadmin@localhost:~$ rm lostfile
sysadmin@localhost:~$ locate lostfile
/home/sysadmin/lostfile
sysadmin@localhost:~$
```

# 3 find Command

- If you want to search for files that are currently in the filesystem, then you should use the **find** command. The **find** command is slower than the **locate** command because it searches directories in real time; however, it doesn't suffer from problems associated with an outdated database.

```
find [OPTIONS]... [starting-point...] [expression]
```

- The **find** command expects a directory as the first argument. This will be the starting point of the search. The **find** command will search this directory and all of its subdirectories. If no directory is specified, then the **find** command will start the search at the current directory.

Refer to the example below on how to use the **find** command

```
sysadmin@localhost:~$ pwd
/home/sysadmin
sysadmin@localhost:~$ find /home/sysadmin -name Downloads
/home/sysadmin/Downloads
sysadmin@localhost:~$ find /home/sysadmin -name Dow*
/home/sysadmin/Downloads
sysadmin@localhost:~$ find $HOME -name Dow*
/home/sysadmin/Downloads
sysadmin@localhost:~$ find ~ -name Dow*
/home/sysadmin/Downloads
sysadmin@localhost:~$ find . -name Dow*
./Downloads
```

Notice that `/home/sysadmin == $HOME == "~"` which may not be `== "."`. `"."` stands for the current directory which in the above example happens to be `/home/sysadmin`. Note also the use of the wildcard `*` which matches zero or occurrences.

### 3 find Command (Cont'd)

- When the user executing the *find* command has limited read privileges, the *find* output will contain errors such as the ones shown below.

```
sysadmin@localhost:~$ find / -name Down*  
find: '/etc/ssl/private': Permission denied  
find: '/root': Permission denied  
find: '/proc/tty/driver': Permission denied  
find: '/proc/1/task/1/fd': Permission denied  
find: '/proc/1/task/1/fdinfo': Permission denied  
find: '/proc/1/task/1/ns': Permission denied  
find: '/proc/1/fd': Permission denied  
find: '/proc/1/map_files': Permission denied  
find: '/proc/1/fdinfo': Permission denied
```

- The *find* command offers many options for searching files. The table on the next slide lists some of the more commonly used ones.

### 3 find Command (Cont'd)

| Option                   | Meaning                                                                          |
|--------------------------|----------------------------------------------------------------------------------|
| <code>-iname</code>      | Case insensitive search by name                                                  |
| <code>-mtime -3</code>   | Files modified less than three days ago.                                         |
| <code>-mmin -10</code>   | Files modified less than ten minutes ago.                                        |
| <code>-size +1M</code>   | Files larger than one megabyte.                                                  |
| <code>-user joe</code>   | Files owned by the user joe.                                                     |
| <code>-nouser</code>     | Files not owned by any user.                                                     |
| <code>-empty</code>      | Files that are empty.                                                            |
| <code>-type d</code>     | Files that are directory files.                                                  |
| <code>-maxdepth 1</code> | Do not use recursion to enter subdirectories; only search the primary directory. |

If multiple criteria are specified, then all criteria must match as the find command automatically assumes a logical **AND** condition between criteria, meaning all of the conditions must be met.

Logical **OR** conditions can be specified between criteria with the **-o** option.

### 3 find Command (Cont'd)

- The following example search within the current directory for files and directories which are either owned by *sysadmin* or has the name *Downloads*.

```
sysadmin@localhost:~$ find . -user sysadmin -o -name Downloads
.
./.profile
./Public
./Templates
./Downloads
./Pictures
./.selected_editor
./.sudo_as_admin_successful
./Desktop
./Documents
```

### 3 find Command (Cont'd)

- To make the output exactly like the output of the `ls -l` command, use the `-exec` option to execute `ls -l` on each file found. For example:

```
sysadmin@localhost:~$ find . -name 'Documents' -exec ls -l {} \;
```

|            |   |          |          |      |             |                  |
|------------|---|----------|----------|------|-------------|------------------|
| total 1100 |   |          |          |      |             |                  |
| drwxr-xr-x | 5 | sysadmin | sysadmin | 4096 | Apr 24 2019 | School           |
| drwxr-xr-x | 2 | sysadmin | sysadmin | 4096 | Apr 24 2019 | Work             |
| -rw-r--r-- | 1 | sysadmin | sysadmin | 39   | Apr 24 2019 | adjectives.txt   |
| -rw-r--r-- | 1 | sysadmin | sysadmin | 90   | Apr 24 2019 | alpha-first.txt  |
| -rw-r--r-- | 1 | sysadmin | sysadmin | 106  | Apr 24 2019 | alpha-second.txt |

- In the above example, the curly braces `{ }` is a placeholder for the name of each file found.

## 3 find Command (Cont'd)

- The following are some other examples of useful ways to use the *find* command.
1. Find files or set of files depending on their age or when they were last modified.
  2. Find files or set of files based on their size

Search for files that have been modified within the last 3 days.

```
sysadmin@localhost:~$ find /tmp -mtime -3  
/tmp
```

Search for files that are at least 20 Kilobytes in size

```
sysadmin@localhost:~$ find /etc/ -size +20k  
/etc/ld.so.cache  
/etc/ssh/moduli  
/etc/mime.types  
find: '/etc/ssl/private': Permission denied  
/etc/ssl/certs/ca-certificates.crt
```

## 4 whereis Command

- There are many commands available to the operating system, and it is sometimes useful to know where they are. The **whereis** command can be used to search your **PATH** for the binary. It is also capable of searching for the man page, and source files for any given command.

```
whereis [OPTION]... NAME...
```

- To find out where the **grep** command is located, use **whereis** without any options:

```
sysadmin@localhost:~$ whereis grep
grep: /bin/grep /usr/share/man/man1/grep.1.gz /usr/share/info/grep.info.gz
```

- The output of the **whereis** command returns three directories. The first is where the **grep** command is located, in the **/bin/grep** directory. There is also a path given for the man page of the **grep** command at **/usr/share/man/man1/grep.1.gz** and another path for the info page at **/usr/share/info/grep.info.gz**. Without an option, the **whereis** command provides all three pieces of information.
- To view only the location of the binary file, use the **-b** option. To view the man and info pages, use the **-m** option.

## 5 which Command

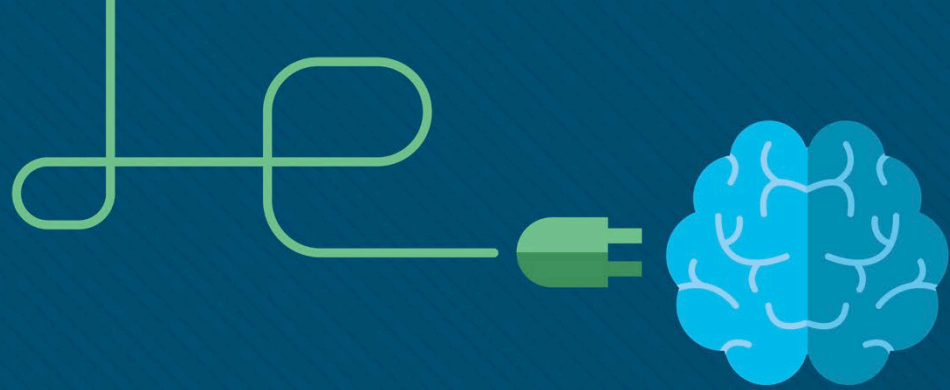
- Sometimes the *whereis* command returns more than one result on a command. In this case, an administrator would want to know which command is being used.

```
which [-a] FILENAME...
```

- The *passwd* command is one of those commands that *whereis* will return more than one result. Use the *which* command to know exactly which file is executed when we run the *passwd* command.

```
sysadmin@localhost:~$ whereis -b passwd
passwd: /usr/bin/passwd /etc/passwd
sysadmin@localhost:~$ which passwd
/usr/bin/passwd
sysadmin@localhost:~$
```

- To find out if there are more than one executable file on the system, use the *-a* option. This helps to identify if malicious versions of existing command files have been inserted to override an existing command.



# Lab

RH124 Chapter 13.7. Lab: Access Linux File System



# Practice Exercises

## 1 Filesystem Hierarchy Standard (FHS)

- List the main directories under / and explain their purposes.
- Navigate to /var/log and list its contents.
- Use ls -l to check the file type and permissions of /etc/passwd.

## 2 Finding Files and Commands

- Use locate to search for bashrc. Why might the results be outdated?
- Update the file database and run locate again.
- Search for all .conf files in /etc using find.
- Find all files modified in the last 3 days.

## 3 Using find for Advanced Searches

- Find all files larger than 5MB in /var/log.
- Locate empty files in /home.
- Search for directories under /usr that do not belong to any user.
- Run find with -exec to list details of found files like ls -l.

## 4 Command Location & Execution

- Use whereis to locate the ls binary and its documentation.
- Use which to determine the exact path of grep.
- Find out if multiple versions of passwd exist on the system.